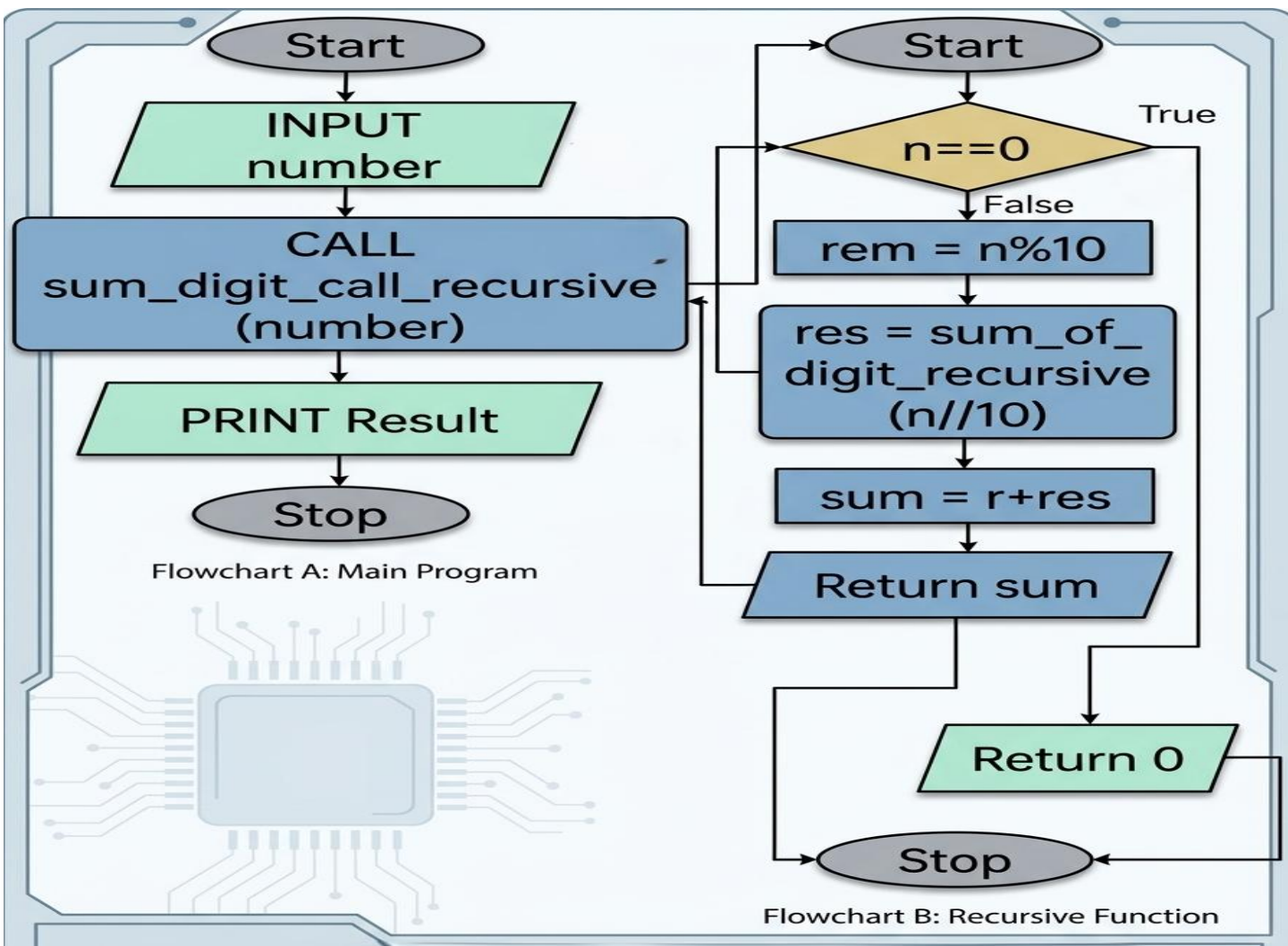
11.1.2 Sum of digits using recursion

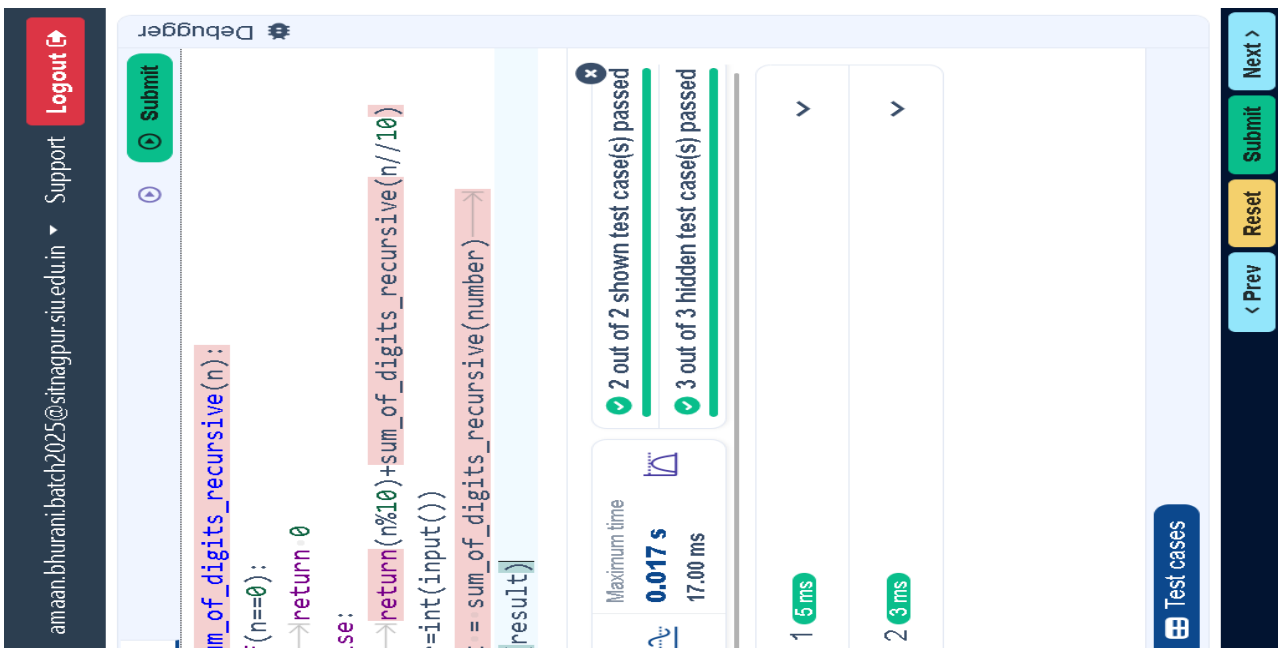
Algorithm :

Step 1: Start
Step 2: Input an integer and store it in number
Step 3: Call the function
sum_of_digits_recursive(number)
Step 4: In the function, check if n == 0
If yes, return 0
Step 5: If no
Find remainder: r = n % 10 (last digit)
Find quotient: q = n // 10 (remaining number)
Step 6: Return r + sum_of_digits_recursive(q)
Step 7: Store the result in result
Step 8: Print result
Step 9: Stop

Flowchart :



Code :

```
def sum_of_digits_recursive(n):
    if n==0:
     return 0
   else:
     return(n%10)+sum_of_digits_recursive(n//10)
number=int(input())
result = sum_of_digits_recursive(number)
print(result)
```

Execution :